

STAT24630

Jingshu Wang

Causal Inference Methods and Case Studies

Lecture 1

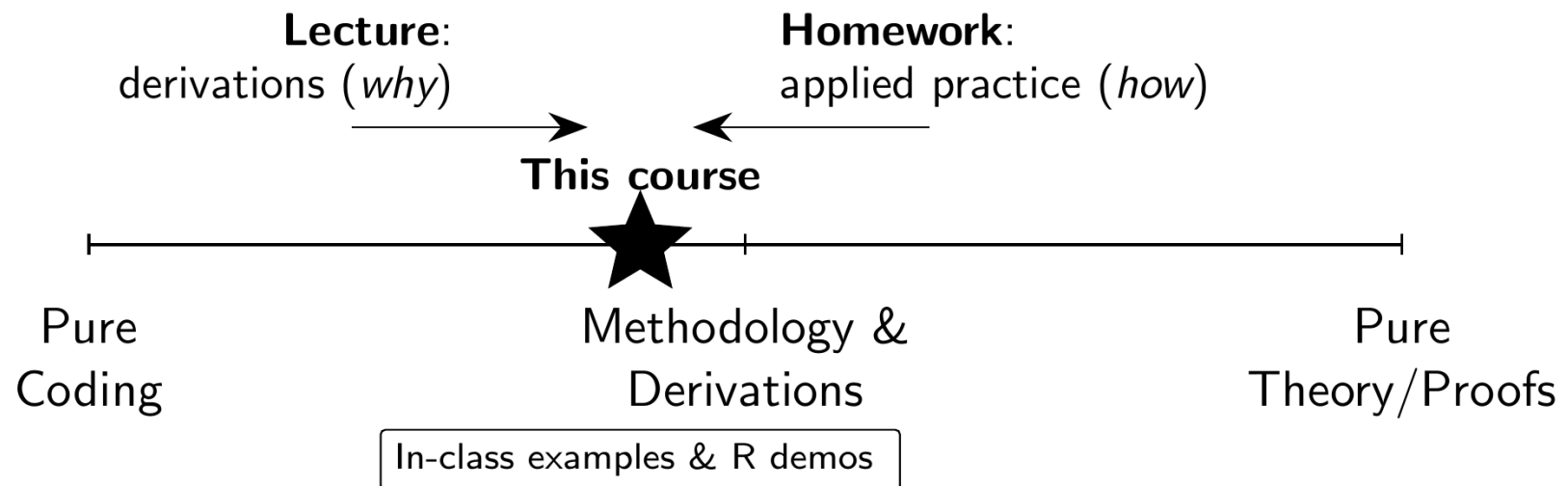
Introduction with four examples

Outline

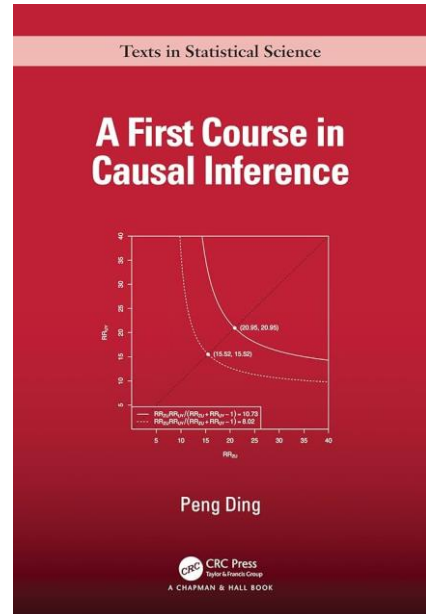
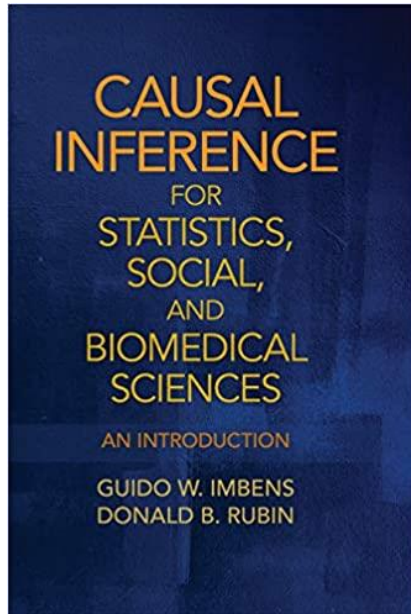
- Course logistics
- HDL: does an association justify an intervention?
- Birthweight: can adjustment create bias?
- Moderna: what does random assignment buy us?
- Schooling: what if we cannot randomize treatment?

What to expect from this course

- Learn causal inference methods and apply them to real data.
- Develop familiarity with causal reasoning, including how to formulate causal questions, set up problems, articulate assumptions, and interpret results in context.
- Use **R or Python** for homework and the project. **Course examples use R.**
- Bring basic programming skills. We will not teach programming fundamentals.
- Lectures include derivations that explain how the methods work.
- Homework combines causal reasoning with data analysis and coding.



Texts and readings



- **Imbens and Rubin (2015)**: Causal Inference for Statistics, Social, and Biomedical Sciences: An Introduction
- [Cambridge Core e-resource](#)
- **Ding (2023)**: A First Course in Causal Inference
- [arXiv e-resource](#)
- Additional articles and case-study materials will be provided.

Assignments and grading

- Four homework assignments: **40%**
- Two in-person quizzes: **20%** (20 minutes each, October 23 and November 18)
- Group project written report: **20%**
- Group presentation and questions: **15%**
- Individual self-assessment: **5%** (1–2 pages, three reflection questions)
- Submit homework via Gradescope by **Sunday, 11:59 PM**, in Weeks 2, 4, 6, and 8.
- Homework deadlines: October 11, October 25, November 8, and November 22.

Group project

- Approximately **four students** per group. Each group uses a **different dataset**.
- Instructor datasets and documentation arrive in **Week 4**. Dataset and causal question approval is due by the end of **Week 6**.
- Proposing your own data requires the dataset, documentation and a preliminary causal question.
- Define the causal question, treatment, and outcome. Apply a course method, assess its assumptions, and explain findings and limitations.
- Presentations on **December 2 and 4**: 10 minutes presenting plus approximately 3 minutes of questions.
- Each member needs a **substantive speaking role**, as a presenter or designated respondent. Submit slides in advance.

Course logistics

- **Class:** Wednesdays and Fridays, 10:30–11:50 AM, Jones 303
- **Instructor:** Jingshu Wang
- **Instructor office hours:** Wednesdays, 2:00–3:00 PM, Jones 317
- **Teaching assistants:** Dong Xie and Qichuan Yin
- **TA office hours:** to be announced
- **Participation:** ask questions, contribute to pair/group activities, discuss case studies, and help peers on Ed.

Course policies

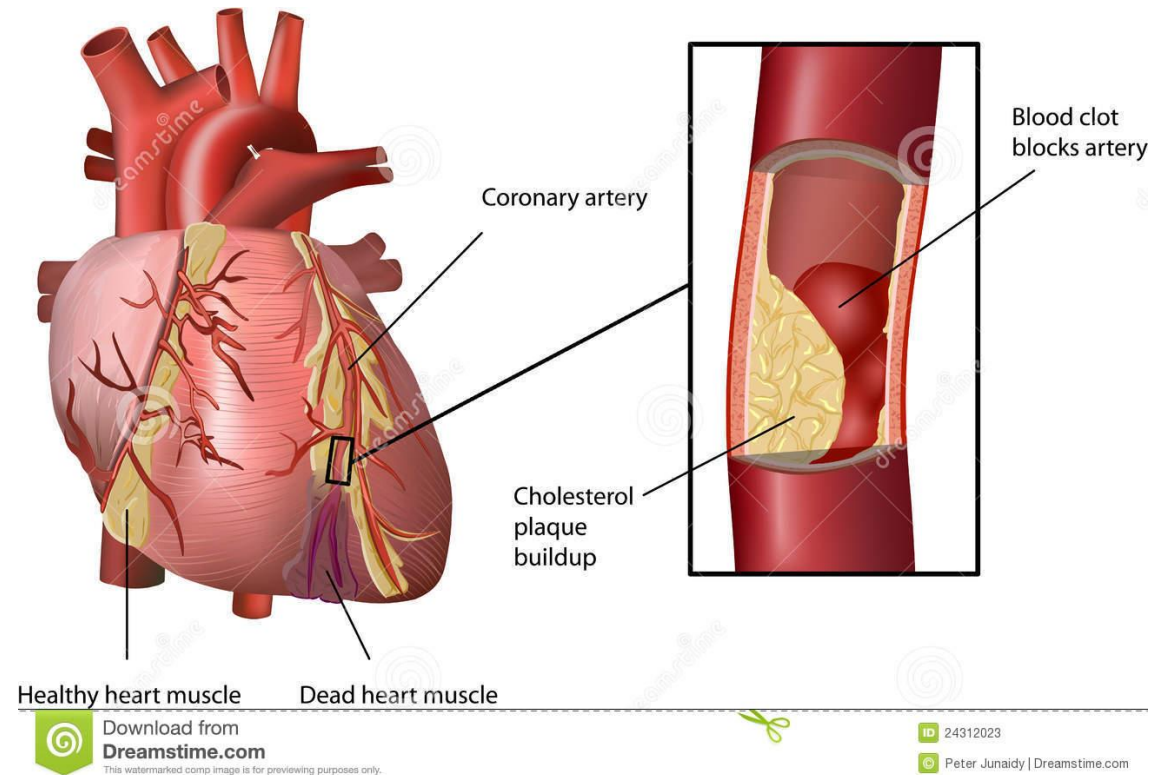
- Homework has a **two-hour grace period** for technical difficulties.
- Late homework and makeup quizzes require **prior approval**, except in documented emergencies.
- Discuss course material and homework with peers after making an honest attempt.
- Write final solutions **independently and in your own words**.
- Complete **your own code and figures**. Comparing final answers to check for mistakes is allowed.
- **Acknowledge substantial help**. Read the syllabus for the full course policies.

Use of AI tools

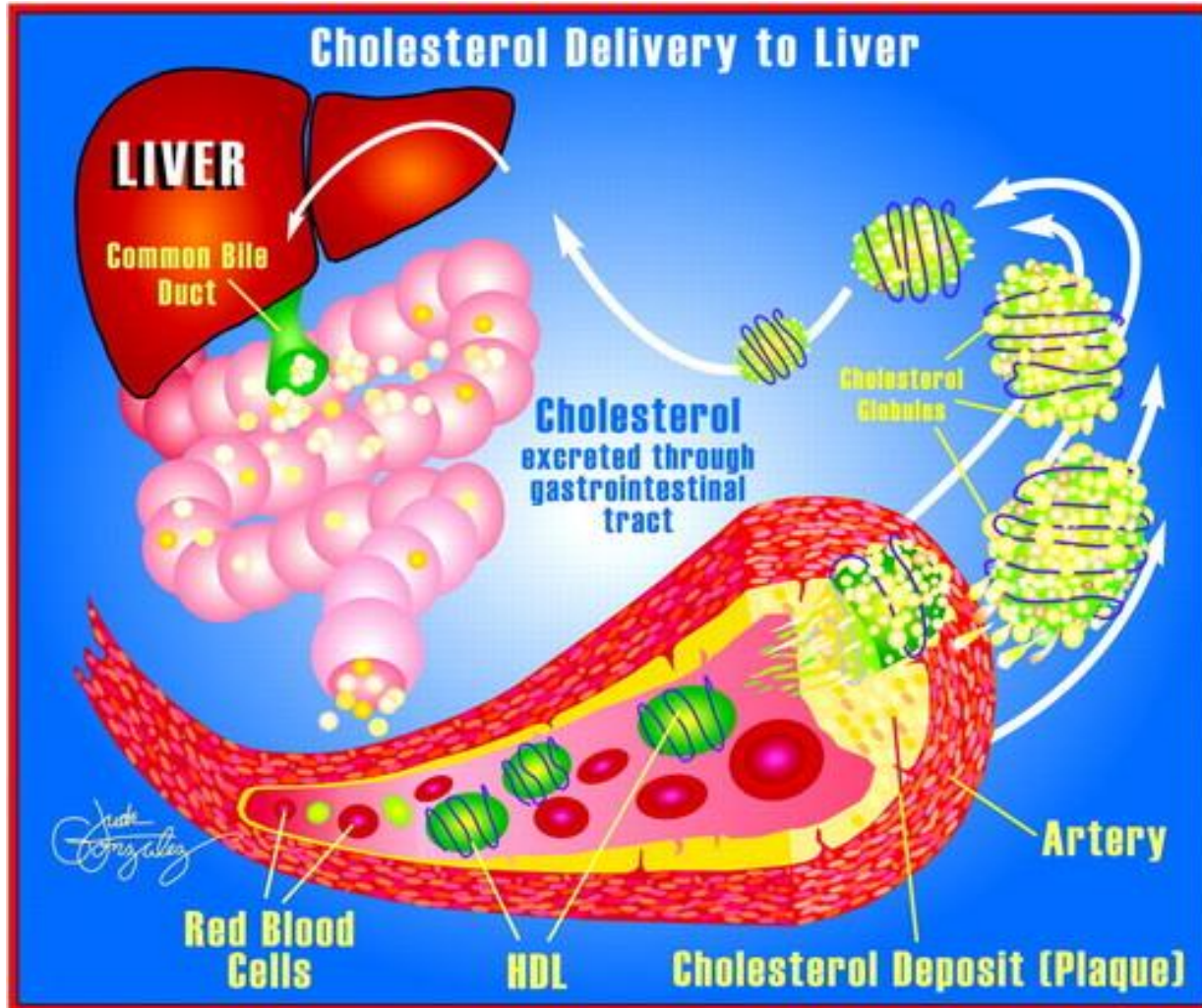
- AI use is optional for homework and the project. AI is **not permitted during quizzes**.
- **Permitted help:** R/Python syntax, debugging, error messages, and grammar or clarity edits without adding substantive content.
- **Verify AI-assisted code** and be able to explain how it works.
- **Your own reasoning** must determine the causal question, identification strategy, assumptions, and interpretation of results.
- AI may not generate complete solutions or substantive explanations.
- **Disclose the tool and its use.** One disclosure in the group report is sufficient. You are responsible for all work and may be asked to explain it.

Example 1

Would a drug that raises HDL-C reduce cardiovascular events?



Why an HDL-raising drug seemed promising



- HDL helps transport cholesterol from tissues back to the liver.
- This biological mechanism motivated a hypothesis: raising HDL-C might reduce cardiovascular events.
- What evidence would justify trying such a drug?

HDL-C and vascular disease: the cohort study

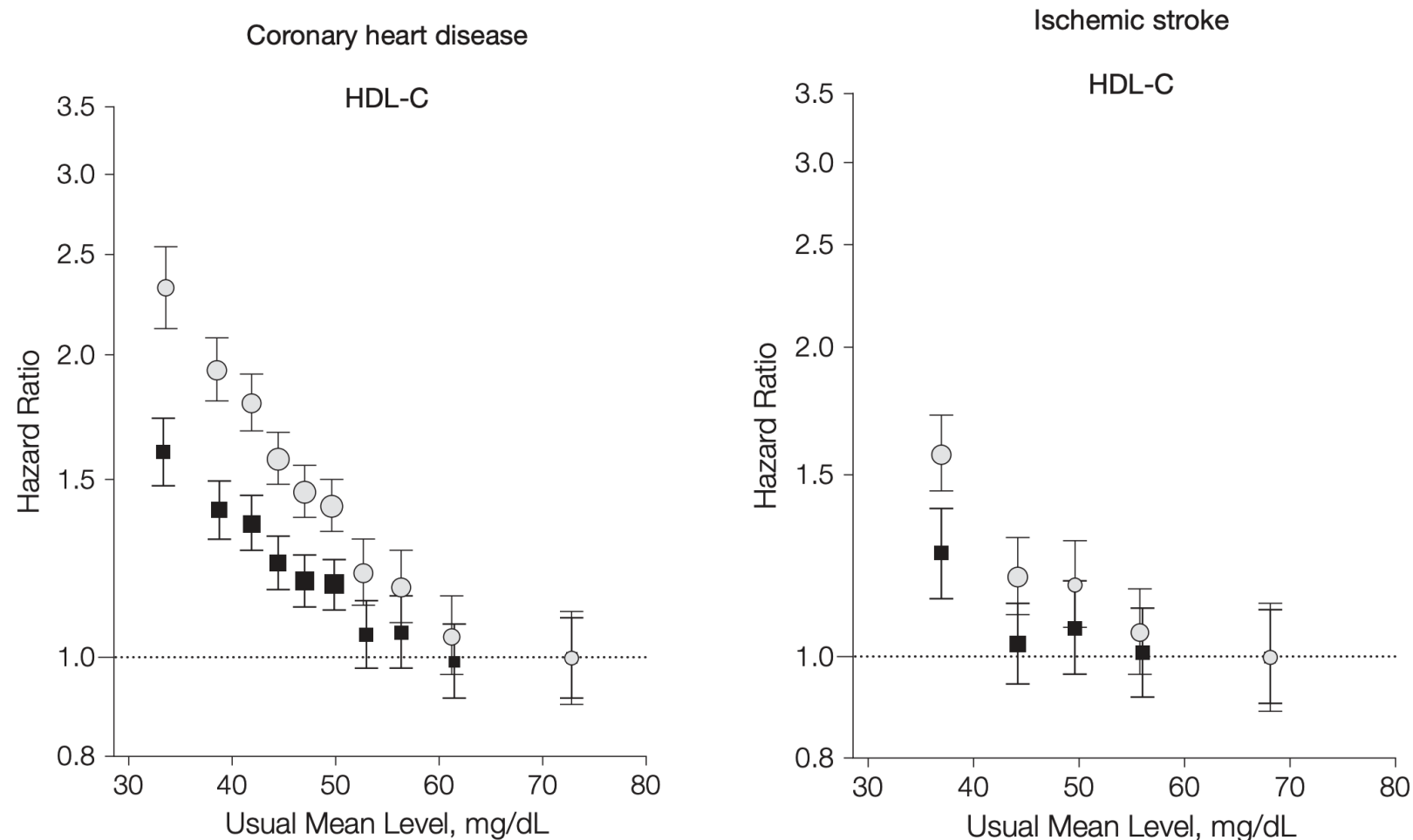
Emerging Risk Factors Collaboration, JAMA (2009)

“Major Lipids, Apolipoproteins, and Risk of Vascular Disease”

- Design: pooled individual records from 68 prospective studies, mostly in Europe and North America.
- Participants: 302,430 people without vascular disease at baseline; 2.79 million person-years of follow-up.
- **Comparison:** Compare the coronary heart disease hazard in each HDL-C group with the hazard in the **highest-HDL-C group**, which serves as the reference.
- **Hazard ratio:** Compares how quickly new cases occur in two groups during follow-up. A ratio above 1 indicates a higher event rate than in the highest-HDL-C reference group.
- **Adjustment:** Account for age, sex, blood pressure, smoking, diabetes, BMI, and other lipid measures. Analyze ischemic stroke separately.

Does following people over time make this comparison causal?

Would you act on this association?



○ Adjusted for age and sex only
■ Further adjusted for several risk factors

- Higher HDL-C is associated with lower CHD hazard.
- Adjustment weakens the association.

Would you develop the drug?

What else could explain the pattern?

A randomized test of an HDL-raising drug

Barter et al., New England Journal of Medicine (2007)

“Effects of Torcetrapib in Patients at High Risk for Coronary Events”

- 15,067 patients at high cardiovascular risk in a randomized, double-blind trial.
- Both groups received atorvastatin, a statin that lowers LDL cholesterol.
- Treatment group also received torcetrapib, an experimental CETP inhibitor that raises HDL cholesterol.
- Comparison: torcetrapib + atorvastatin versus atorvastatin alone.
- Primary outcome: time to the first major cardiovascular event.

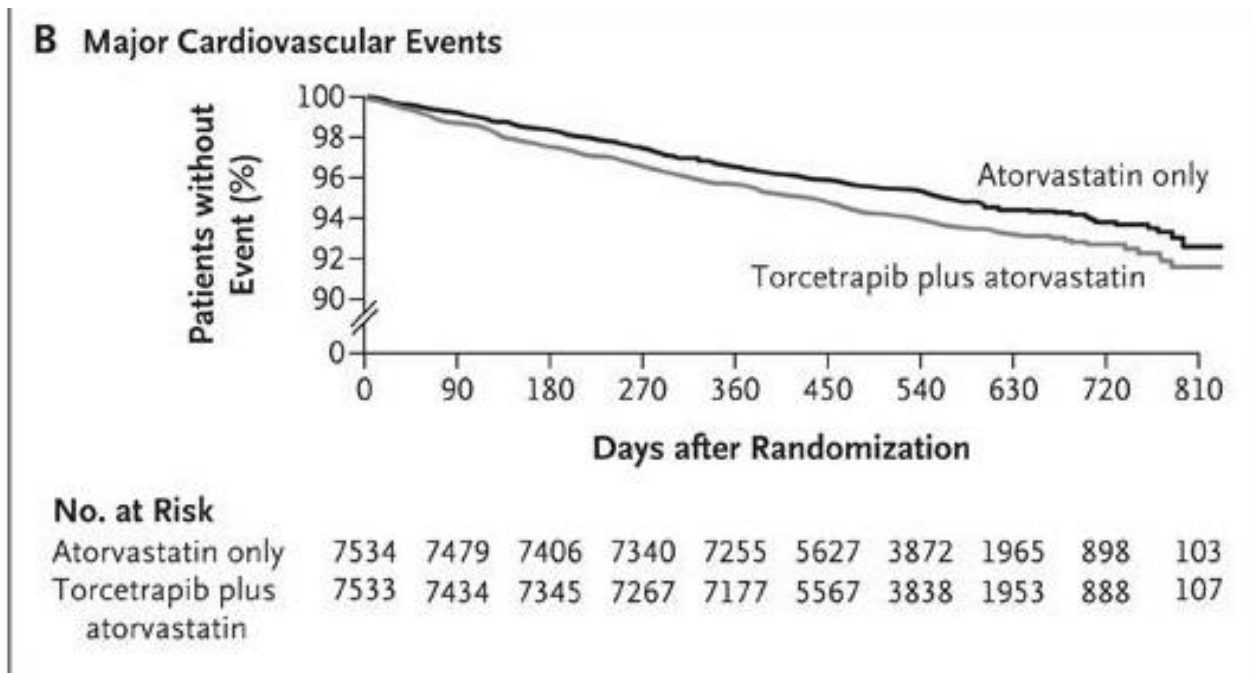
Prediction: which group should have fewer cardiovascular events?

The drug raised HDL-C but did not help

Table 2. Changes from Baseline at 3 Months and 12 Months in Selected Measures.*

Variable	Change at 3 Months			Change at 12 Months		
	Atorvastatin Only	Torcetrapib plus Atorvastatin	P Value	Atorvastatin Only	Torcetrapib plus Atorvastatin	P Value
Lipids (absolute change) — mg/dl						
Cholesterol						
Total	+1.6±20.5	+5.1±23.9	<0.001	+2.1±22.4	+9.3±26.3	<0.001
High-density lipoprotein	+0.5±6.2	+29.0±14.4	<0.001	+0.5±6.8	+34.2±17.0	<0.001
Low-density lipoprotein	+0.6±15.8	-20.5±20.8	<0.001	+0.9±17.1	-21.5±22.7	<0.001

- HDL-C increased substantially.
- The torcetrapib group had more major cardiovascular events.
- The trial estimates this drug regimen's effect, not the effect of HDL-C alone.



Do the cohort study and the trial answer the same question?

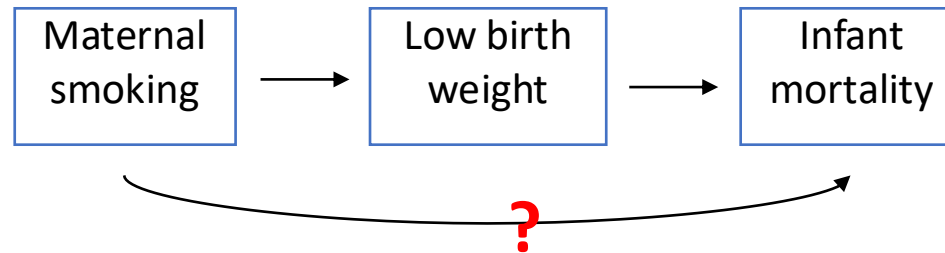
Example 2

Why can smoking appear protective among low-birthweight babies?



What effect of maternal smoking do we mean?

- Smoking is associated with lower birthweight. Low birthweight predicts mortality.
- Total effect: how would mortality change if mothers did not smoke?
- Direct effect: are there effects through pathways other than birthweight?

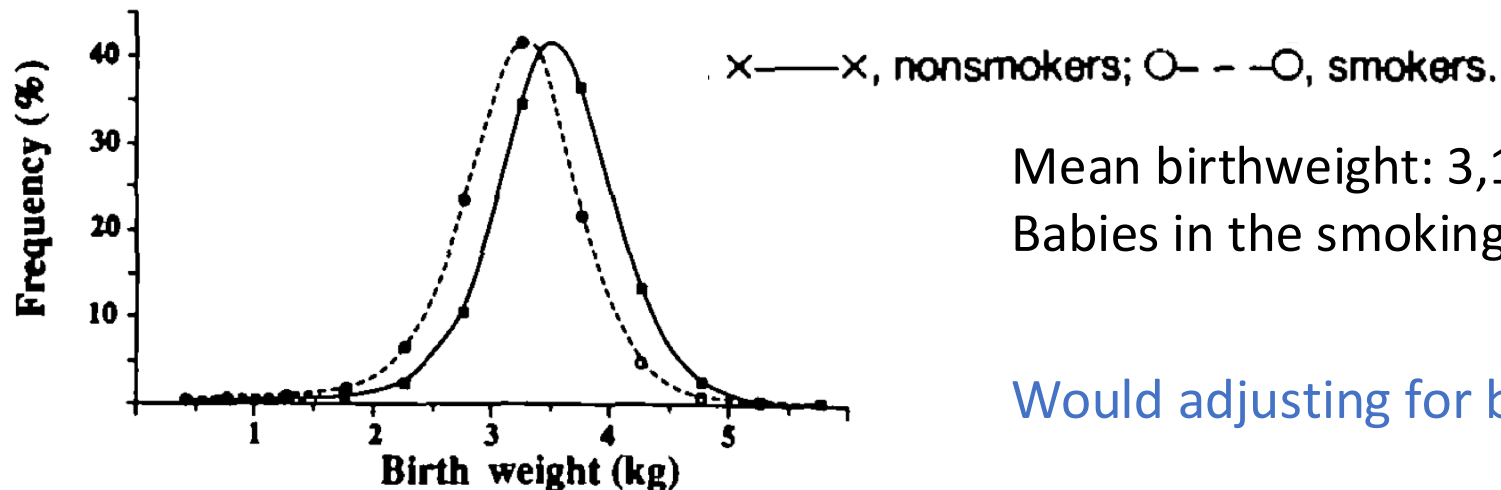


Would comparing babies of the same birthweight answer the direct-effect question?

Smoking and mortality: the overall comparison

[Wilcox \(1993\). American Journal of Epidemiology 137:1098–1104.](#)

- Missouri vital records, 1980–1984. Observational comparison of maternal smoking during pregnancy.
- 215,428 births with no reported smoking, versus 42,270 with at least one pack per day.
- Perinatal deaths: stillbirths plus deaths in the first 28 days.
- Mortality: 14.5 per 1,000 in the smoking group, versus 10.4 per 1,000 in the nonsmoking group.



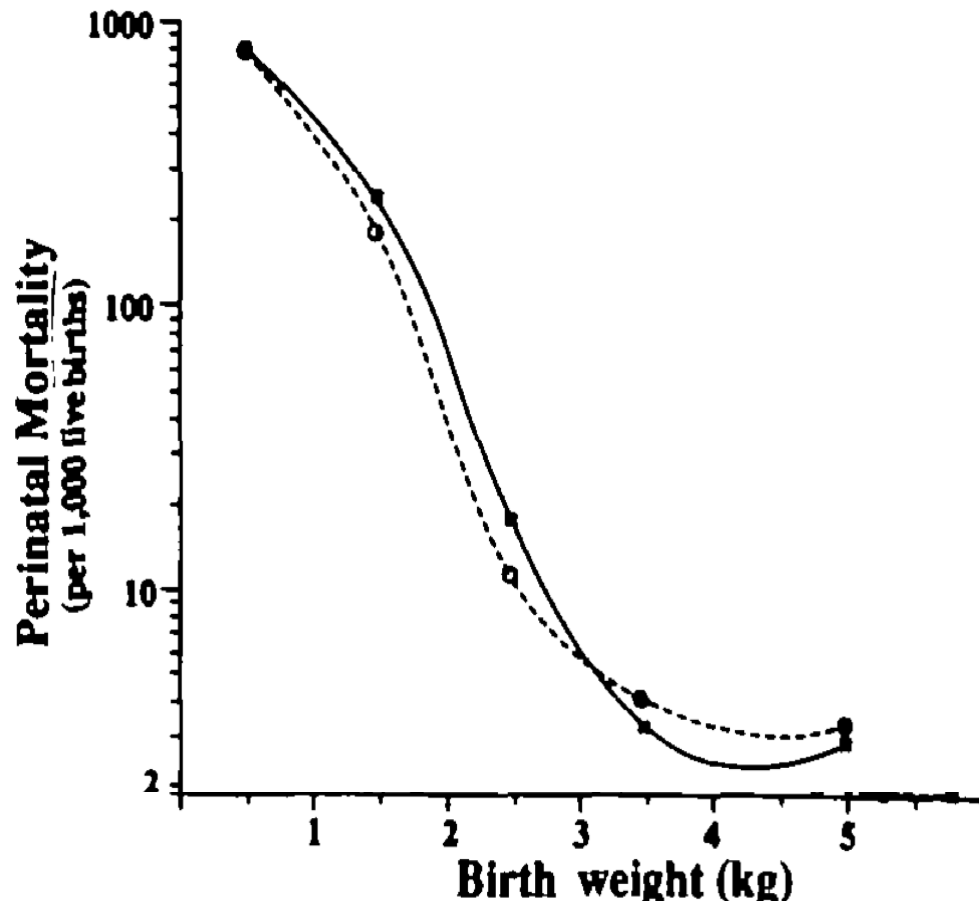
Mean birthweight: 3,180 g versus 3,500 g.
Babies in the smoking group were 320 g lighter.

Would adjusting for birthweight help?

The comparison reverses within birthweight

[Wilcox \(1993\). American Journal of Epidemiology 137:1098–1104.](#)

×—×, nonsmokers; ○— —○, smokers.



Below about 3 kg, babies in the smoking group have lower mortality than babies of nonsmokers at the same birthweight.

Two babies have the same low birthweight.
What else might differ between them?

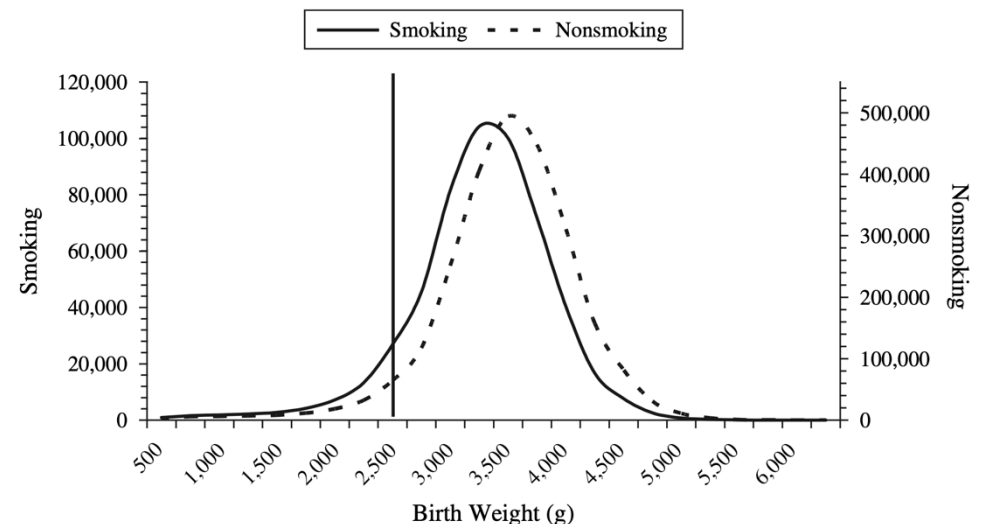
A similar pattern in a second population

[Hernández-Díaz, Schisterman & Hernán \(2006\). American Journal of Epidemiology 164:1115–1120.](#)

- About 3 million U.S. live births in 1991, linked to infant-death records.
- Comparison: babies of mothers who reported smoking during pregnancy versus those who did not.
- Outcome: death before age 1. Low birthweight means less than 2,500 g.
- Analysis: compare mortality within 250 g birthweight categories, with adjustment for maternal characteristics.

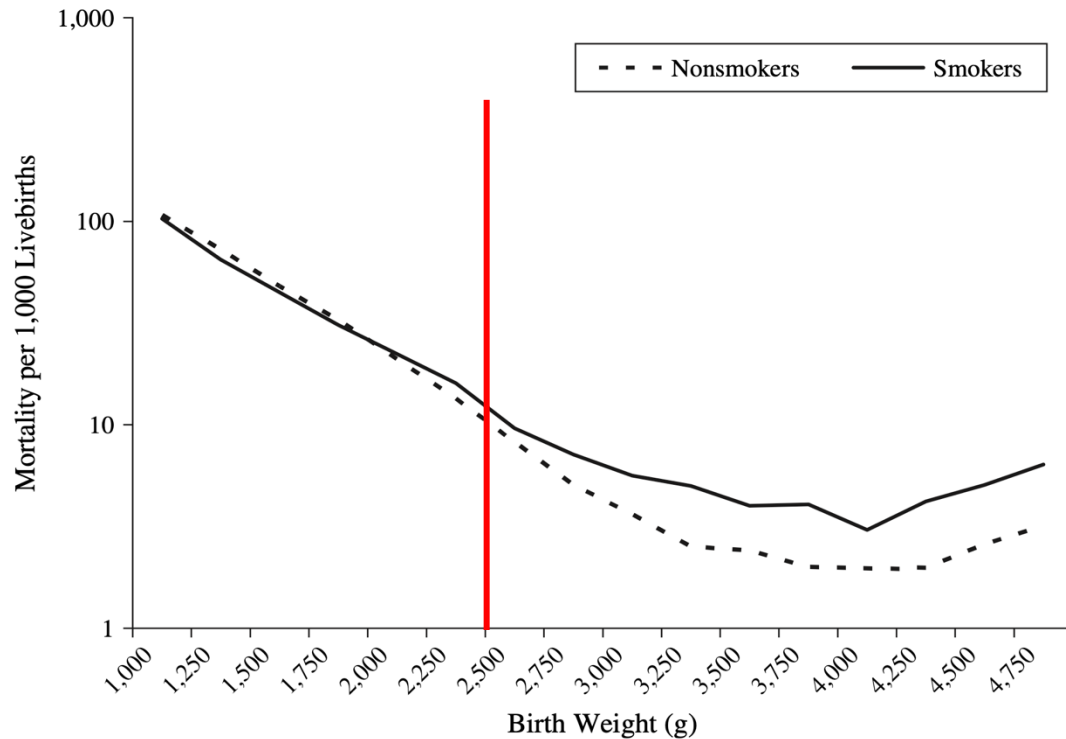
Mean birthweight: 3,145 g with smoking
versus 3,370 g without smoking.

A larger sample does not repair
a biased comparison.



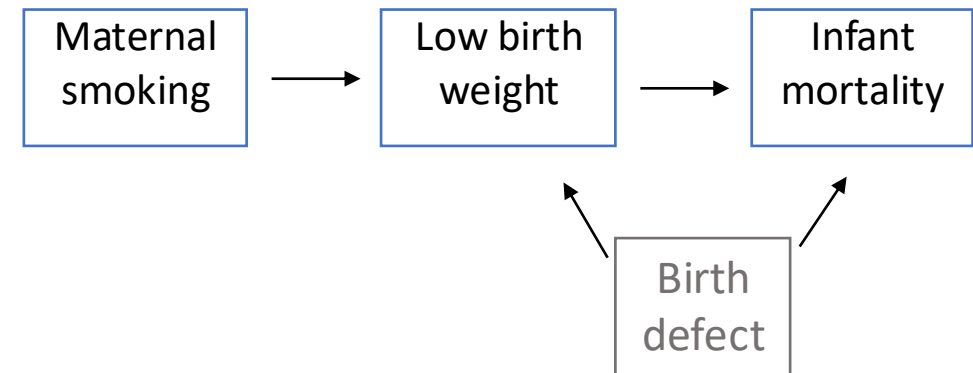
Conditioning can create a misleading association

[Hernández-Díaz, Schisterman & Hernán \(2006\). American Journal of Epidemiology 164:1115–1120.](#)



Mortality rate ratio: smoking / no smoking
Below 2,500 g: 0.79 (95% CI 0.76–0.82).
At least 2,500 g: 1.80 (95% CI 1.72–1.88).

A possible explanation: collider bias



- Smoking and other causes can both lower birthweight.
- Within low birthweight, babies of nonsmokers may have more of the other harmful causes.
- Adjusting for birthweight can introduce bias because smoking and other harmful conditions both affect birthweight.

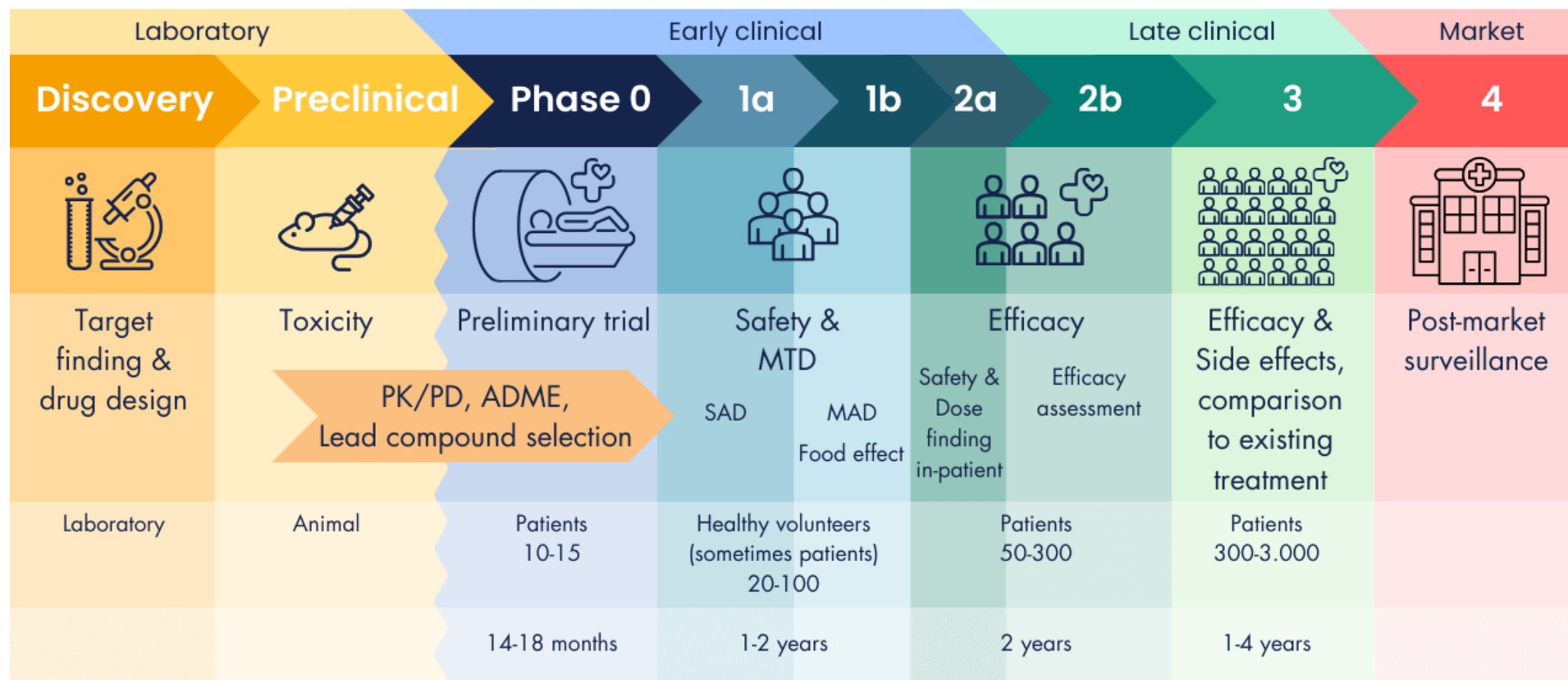
Example 3

How can we create a
credible comparison?
The Moderna vaccine trial



Where randomized trials fit in drug development

Phases of drug development



PK Pharmacokinetics
PD Pharmacodynamics

ADME Absorption, Distribution, Metabolism and Excretion
MTD Maximum Tolerated Dose

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Simplified representation of phases in drug development. Study content and dates and numbers given may vary between studies. No rights can be derived from this figure.

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Clinical trial phases

- Phase 0: exploratory studies of very small doses and early biological activity.
- Phase 1: safety, dose, and what happens to the treatment in the body.
- Phase 2: preliminary efficacy and further safety evaluation.
- Phase 3: larger studies comparing efficacy and safety with a relevant control.
- Phase 4: studies after approval, including longer-term benefits and risks.

Phase describes the study's purpose. Randomization describes its design.

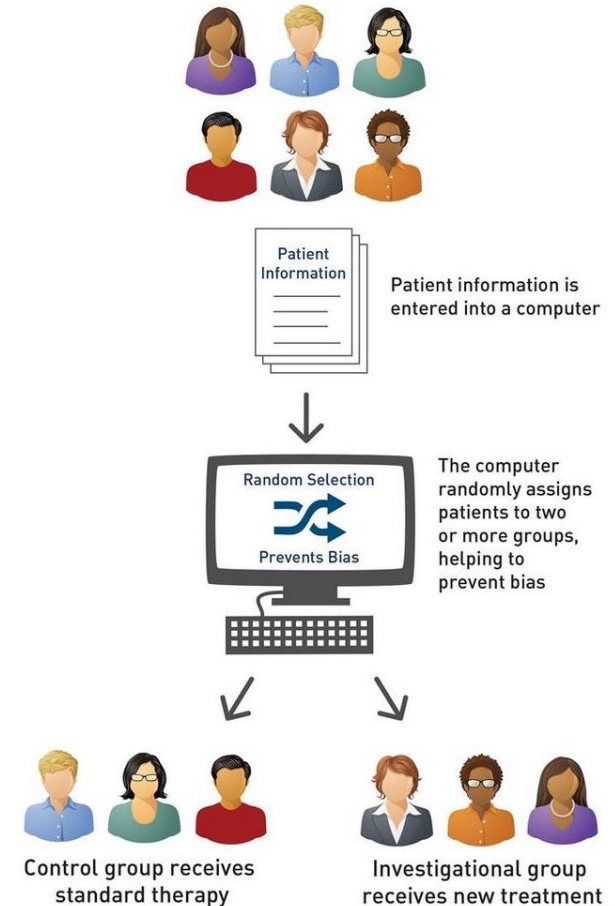
[Phase overview: FDA, "Step 3: Clinical Research"](#)

<https://www.ipmglobal.org/our-work/research/clinical-trial/clinical-trials-approach>

Why random assignment helps

Why might people who choose vaccination differ from those who do not?

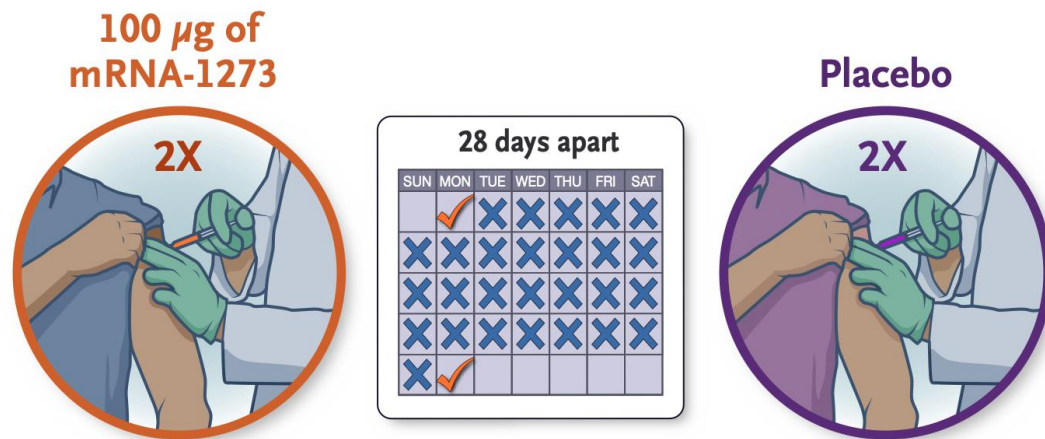
- Random assignment breaks the systematic link between treatment assignment and baseline causes of the outcome.
- The placebo group helps estimate what would happen without the vaccine.



The Moderna trial: treatment, comparison, outcome

[Baden et al. \(2021, published online 2020\). New England Journal of Medicine 384:403–416.](#)

- Phase 3 trial: 30,420 adults at 99 U.S. centers in 2020, randomized 1:1 (15,210 per group).
- Treatment: two 100 µg doses of mRNA-1273. Comparison: two placebo injections, 28 days apart.
- Observer-blinded. Primary outcome: symptomatic, test-confirmed COVID-19 starting 14 days after dose 2.
- Median follow-up: 64 days after dose 2 at the November 25, 2020 analysis cutoff.



mRNA-1273 teaches the immune system to recognize the SARS-CoV-2 spike protein.

Randomization and baseline balance

[Baden et al. \(2021, published online 2020\). New England Journal of Medicine 384:403–416.](#)

Randomization balances baseline characteristics on average.
Chance imbalances can remain in a particular trial.

Table 1. Demographic and Clinical Characteristics at Baseline.*

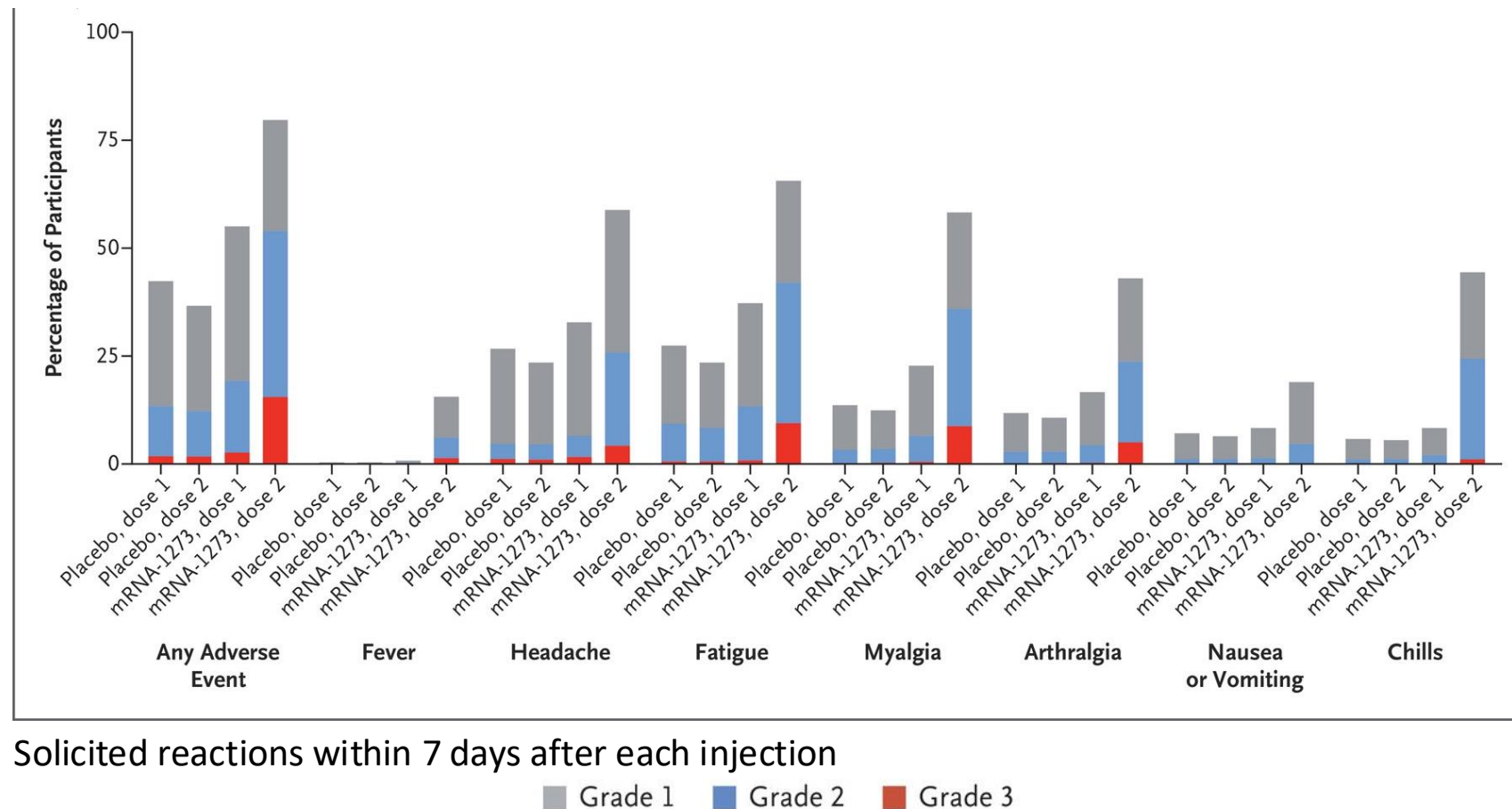
Characteristics	Placebo (N=15,170)	mRNA-1273 (N=15,181)	Total (N=30,351)
Sex — no. of participants (%)			
Male	8,062 (53.1)	7,923 (52.2)	15,985 (52.7)
Female	7,108 (46.9)	7,258 (47.8)	14,366 (47.3)
Mean age (range) — yr	51.3 (18–95)	51.4 (18–95)	51.4 (18–95)
Age category and risk for severe Covid-19 — no. of participants (%)†			
18 to <65 yr, not at risk	8,886 (58.6)	8,888 (58.5)	17,774 (58.6)
18 to <65 yr, at risk	2,535 (16.7)	2,530 (16.7)	5,065 (16.7)
≥65 yr	3,749 (24.7)	3,763 (24.8)	7,512 (24.8)
Hispanic or Latino ethnicity — no. of participants (%)‡			
Hispanic or Latino	3,114 (20.5)	3,121 (20.6)	6,235 (20.5)
Not Hispanic or Latino	11,917 (78.6)	11,918 (78.5)	23,835 (78.5)
Not reported and unknown	139 (0.9)	142 (0.9)	281 (0.9)
Race or ethnic group — no. of participants (%)‡			
White	11,995 (79.1)	12,029 (79.2)	24,024 (79.2)
Black or African American	1,527 (10.1)	1,563 (10.3)	3,090 (10.2)
Asian	731 (4.8)	651 (4.3)	1,382 (4.6)
American Indian or Alaska Native	121 (0.8)	112 (0.7)	233 (0.8)
Native Hawaiian or Other Pacific Islander	32 (0.2)	35 (0.2)	67 (0.2)

Adverse events need a comparison group

[Baden et al. \(2021, published online 2020\). New England Journal of Medicine 384:403–416.](#)

A participant has a headache after vaccination.

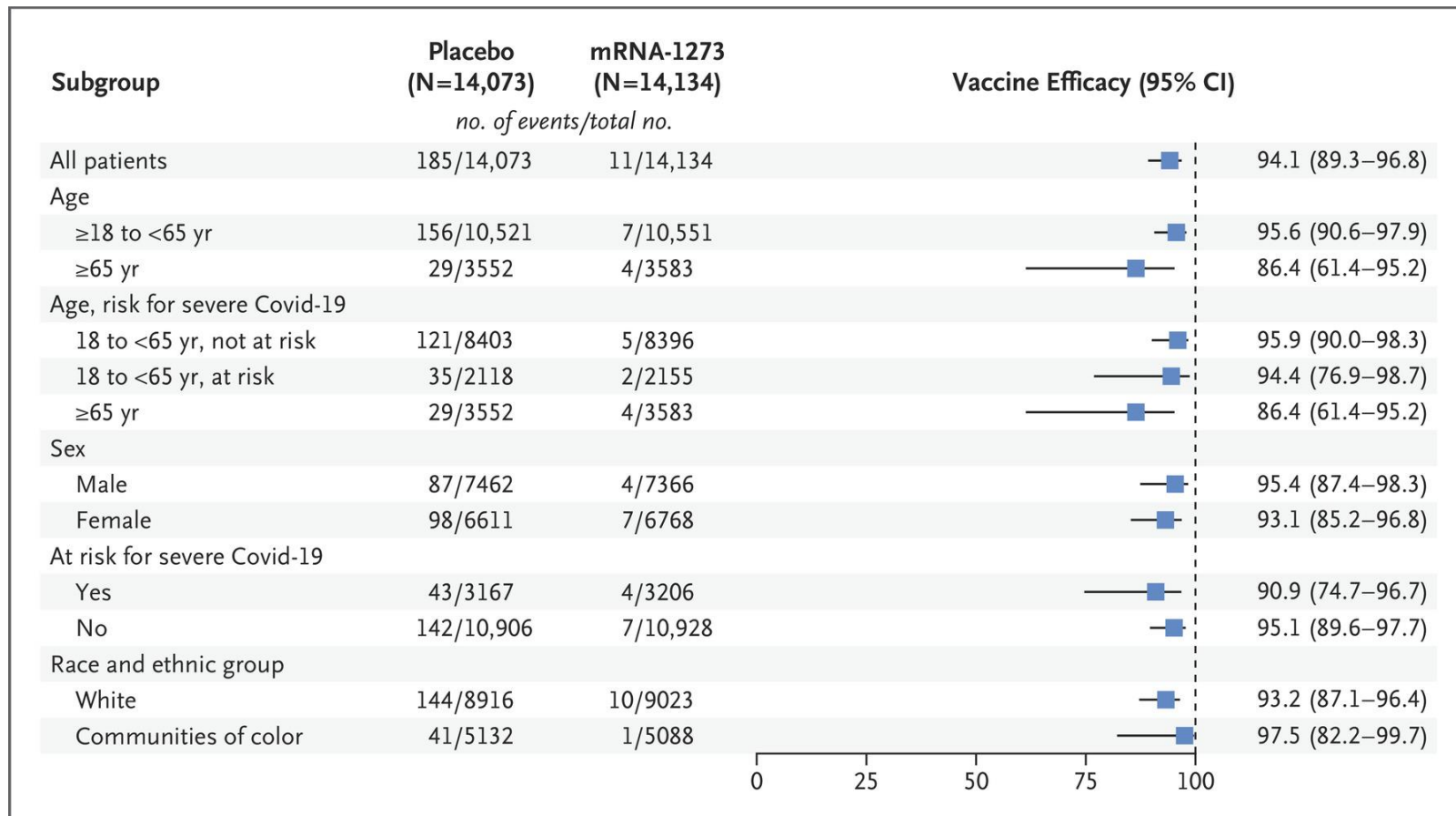
What comparison helps us assess whether the vaccine caused it?



Moderna: efficacy and uncertainty

Primary analysis: 11 / 14,134 vaccine recipients versus 185 / 14,073 placebo recipients.
Reported efficacy: 94.1% (95% CI 89.3–96.8%).

Risk-based approximation: $1 - (11 / 14,134) / (185 / 14,073) \approx 94.1\%$

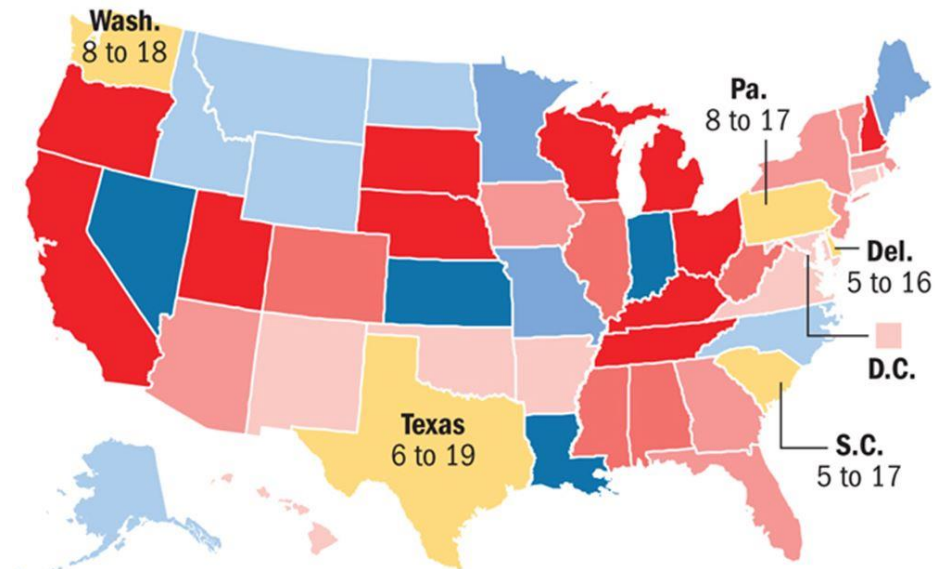


Example 4

Does additional schooling
increase later earnings?

Compulsory school attendance laws, minimum and maximum age limits

AGE OF REQUIRED SCHOOL ATTENDANCE, 2017



Note: For specifics regarding each state's law, go to nces.ed.gov

Source: National Center for Education Statistics

Post-Gazette

Schooling and earnings: the confounding problem

[Angrist & Krueger \(1991\): “Does Compulsory School Attendance Affect Schooling and Earnings?”](#)

- People with more schooling tend to earn more.
- Ability and family background may affect both schooling and earnings.

$$\log(\text{Weekly wage}_i) = \beta_0 + \beta_1 \text{Schooling}_i + \beta_2 \text{Ability}_i + \text{noise}_i$$

Omitting ability can bias the estimated schooling effect when ability predicts both schooling and earnings.

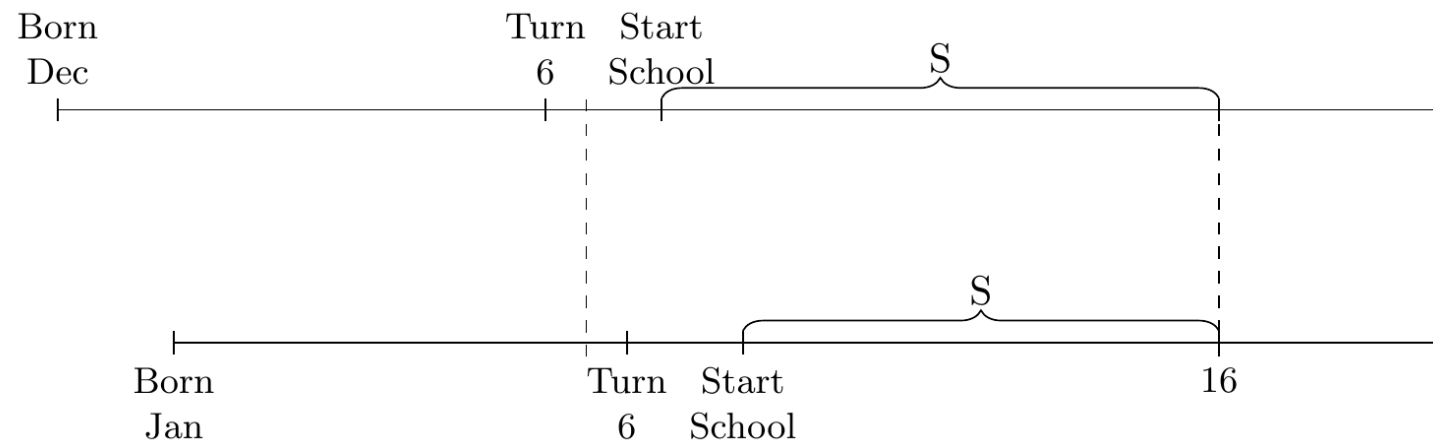
Can something change schooling without reflecting earning potential?

School-entry rules and the legal dropout age

[Angrist & Krueger \(1991\). Quarterly Journal of Economics 106:979–1014.](#)

- Historical illustration: school entry requires age 6 by January 1 of the entry year.
- Children may leave school when they reach age 16.

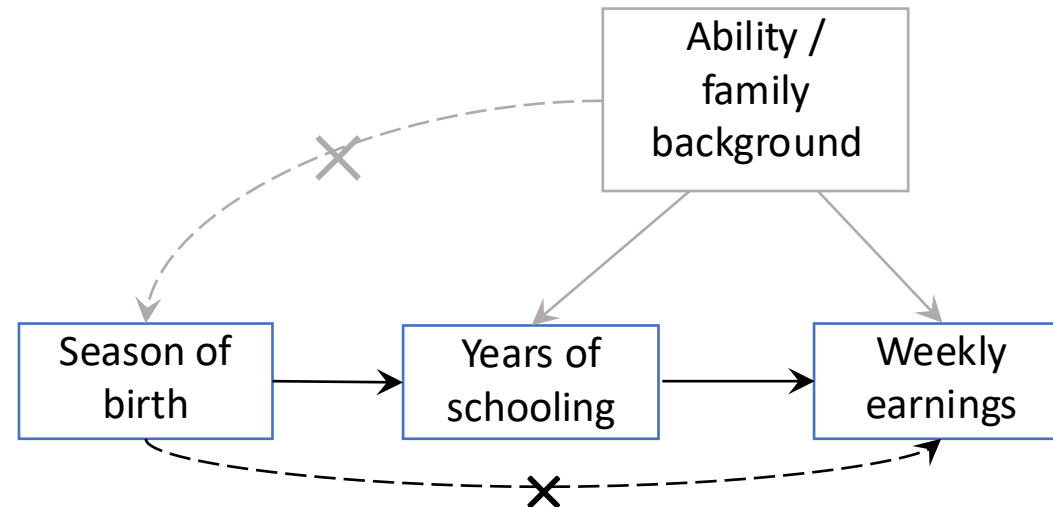
Who can leave school with fewer completed years: the January-born child or the December-born child?



Season of birth as an instrument

[Angrist & Krueger \(1991\). Quarterly Journal of Economics 106:979–1014.](#)

- Birth quarter predicts schooling through entry rules and attendance laws.
- For birth quarter to be a valid instrument, it must be unrelated to other determinants of earnings and affect earnings only through schooling.



Birth quarter predicts years of schooling

Study data: U.S. Census records. The following figures show men born in 1930–1949.

[Angrist & Krueger \(1991\). Quarterly Journal of Economics 106:979–1014.](#)

Within birth cohorts, earlier birth quarters tend to have less schooling.

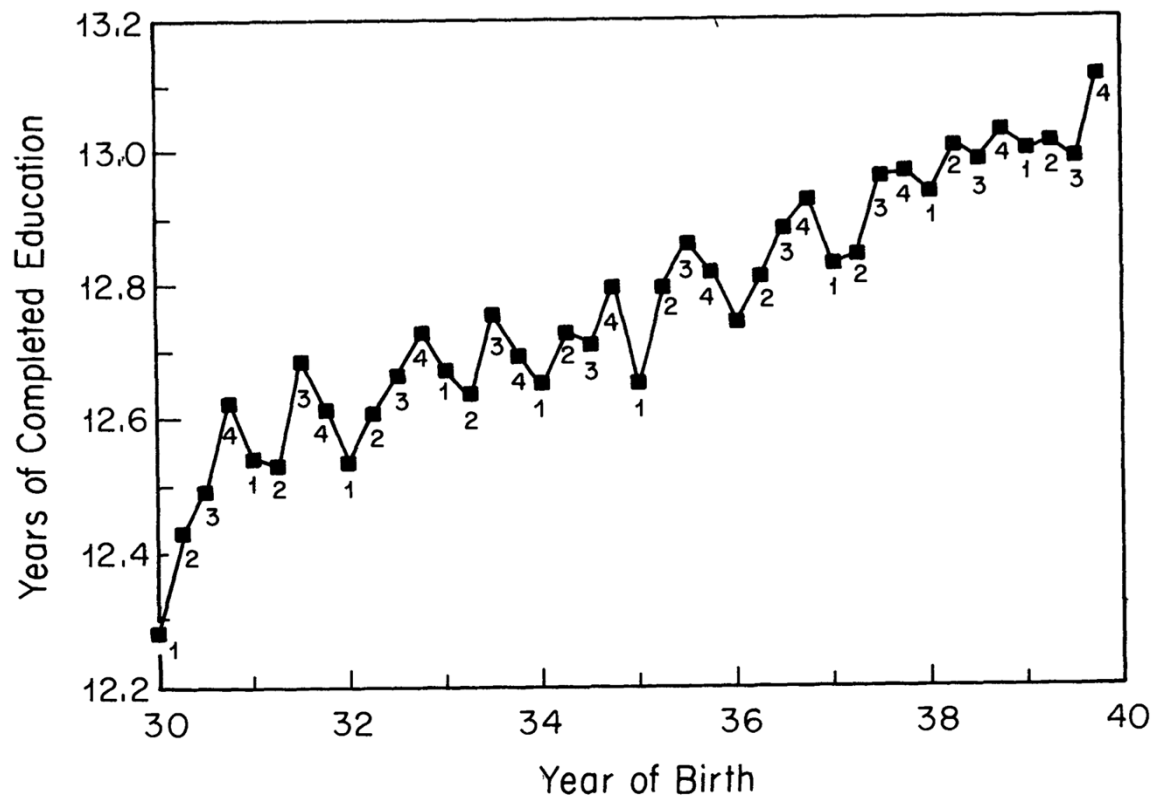
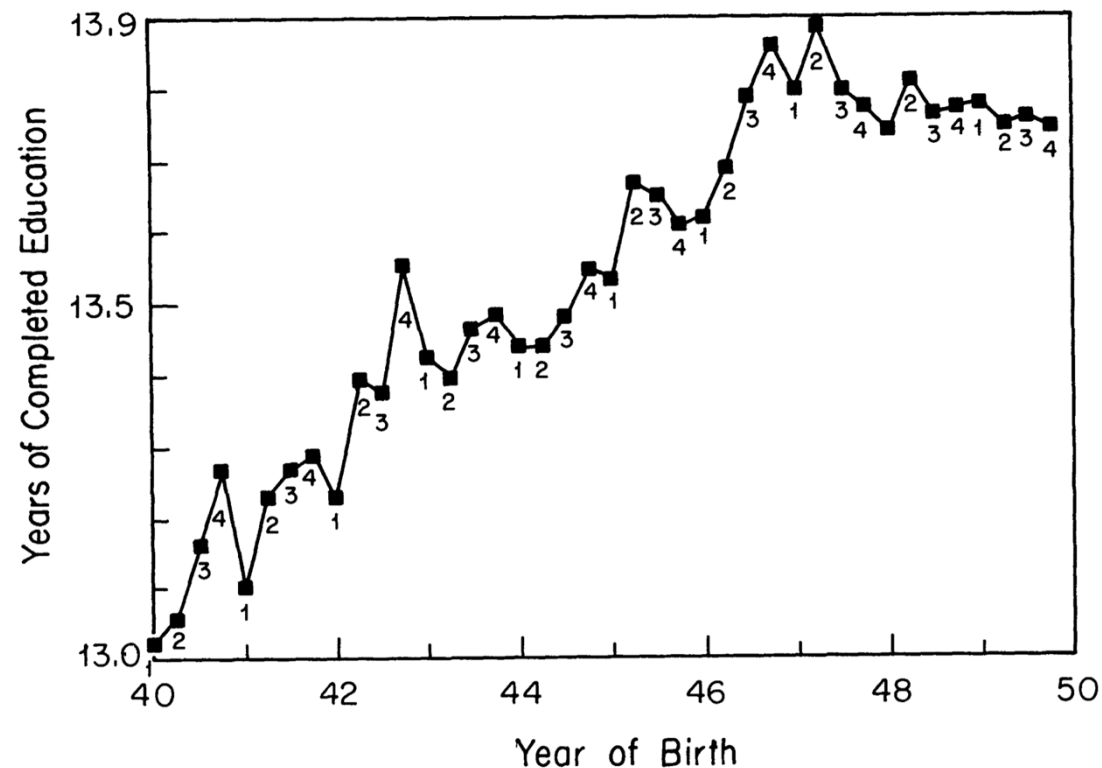


FIGURE I
Years of Education and Season of Birth
1980 Census
Note. Quarter of birth is listed below each observation.

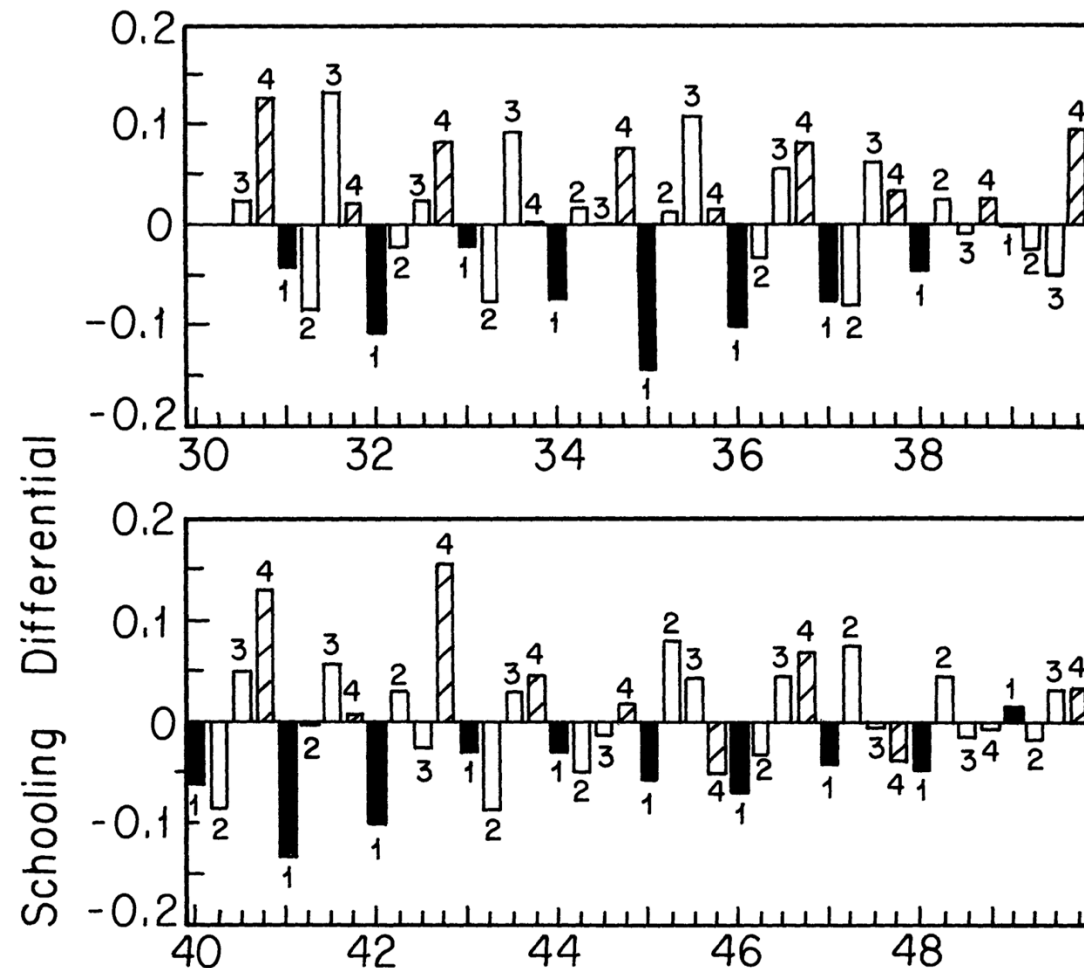


1980 U.S. Census, men born 1930–1949.
Quarter 1 = Jan–Mar; quarter 4 = Oct–Dec.

The pattern after removing the year trend

[Angrist & Krueger \(1991\). Quarterly Journal of Economics 106:979–1014.](#)

After removing the cohort trend, compare quarters within each birth year.



Birth quarter and earnings

1980 Census, men born 1930–1949: earlier birth quarters tend to have lower earnings.

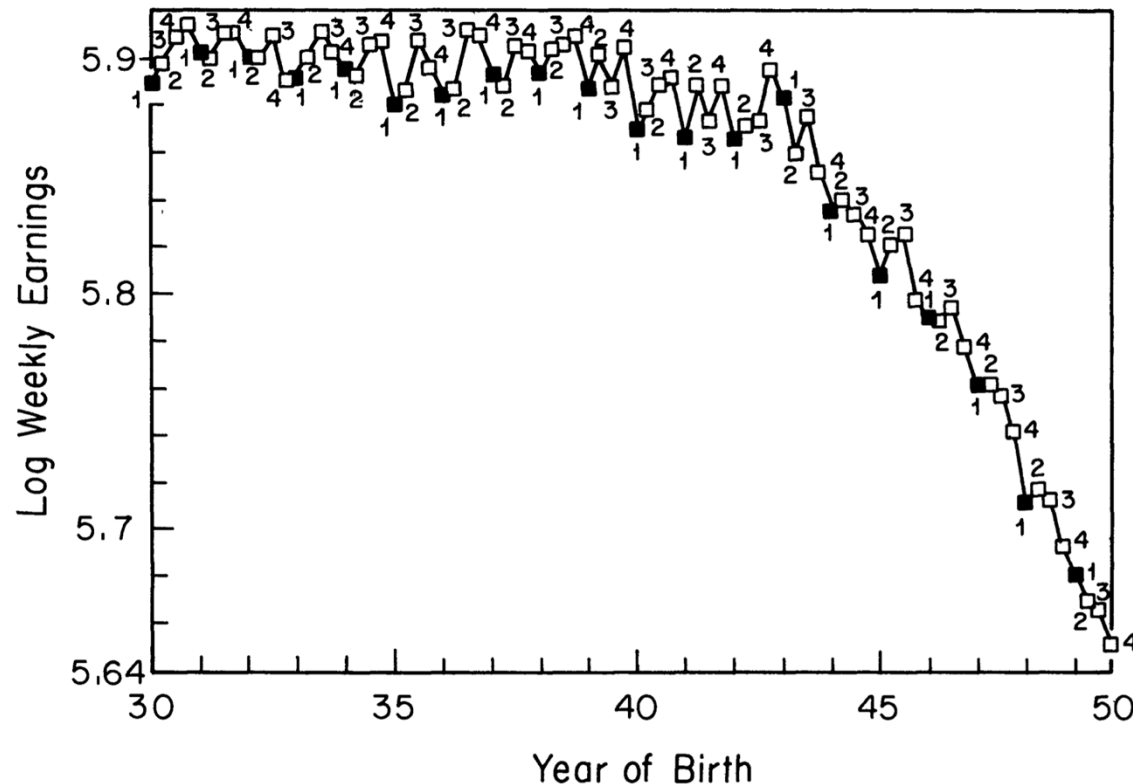


FIGURE V
Mean Log Weekly Wage, by Quarter of Birth
All Men Born 1930–1949; 1980 Census

The causal interpretation requires the earnings difference to operate through schooling.

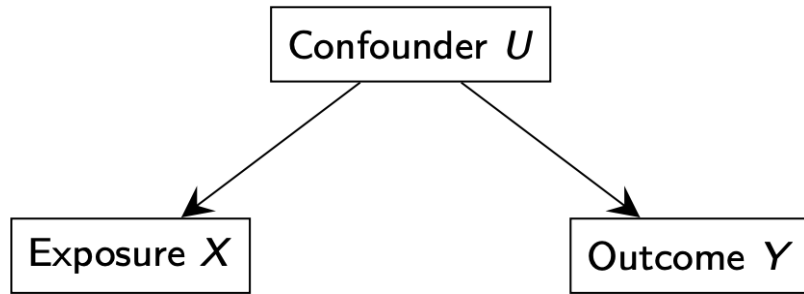
Challenge the design:
Name one way birth quarter could affect earnings without changing years of schooling.

What each example taught us

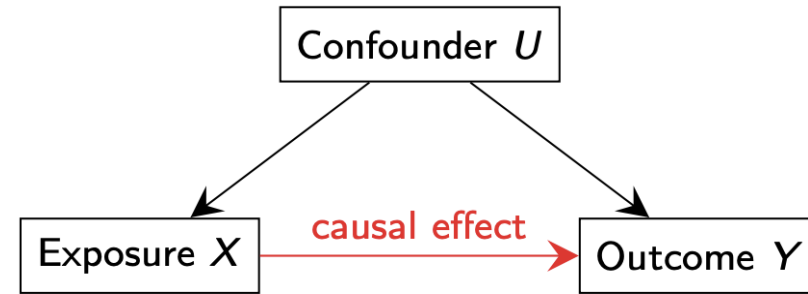
- HDL: an association does not isolate the effect of a particular intervention.
- Birthweight: conditioning can create a misleading association.
- Moderna: random assignment provides a credible comparison in expectation.
- Schooling: an instrument requires substantive assumptions beyond a visible association.

In every example: what comparison supports the causal claim?

A common framework for causal questions



(a) Correlated but not causal



(b) Causal

- What intervention, outcome, population, and time period do we mean?
- Which groups are compared, and why might they differ?
- What design or assumptions justify interpreting the comparison causally?

Reference papers to read

Example 1:

- Emerging Risk Factors Collaboration (2009). Major lipids, apolipoproteins, and risk of vascular disease. *JAMA*, 302(18), 1993-2000. <https://jamanetwork.com/journals/jama/article-abstract/184863>
- Barter, P. J., Caulfield, M., Eriksson, M., Grundy, S. M., Kastelein, J. J., Komajda, M., ... & Brewer, B. (2007). Effects of torcetrapib in patients at high risk for coronary events. *New England journal of medicine*, 357(21), 2109-2122. https://www.nejm.org/doi/10.1056/NEJMoa0706628?url_ver=Z39.88-2003&rfr_id=ori:rid:crossref.org&rfr_dat=cr_pub%20%200www.ncbi.nlm.nih.gov

Example 2:

- Wilcox, A. J. (1993). Birth weight and perinatal mortality: the effect of maternal smoking. *American journal of epidemiology*, 137(10), 1098-1104. <https://academic.oup.com/aje/article-abstract/137/10/1098/128195>
- Hernández-Díaz, S., Schisterman, E. F., & Hernán, M. A. (2006). The birth weight “paradox” uncovered?. *American journal of epidemiology*, 164(11), 1115-1120. <https://academic.oup.com/aje/article/164/11/1115/61454>

Example 3:

- Baden, L. R., El Sahly, H. M., Essink, B., Kotloff, K., Frey, S., Novak, R., ... & Zaks, T. (2020). Efficacy and safety of the mRNA-1273 SARS-CoV-2 vaccine. *New England journal of medicine*. <https://www.nejm.org/doi/full/10.1056/nejmoa2035389>

Example 4:

- Angrist, J. D., & Krueger, A. B. (1991). Does compulsory school attendance affect schooling and earnings?. *The Quarterly Journal of Economics*, 106(4), 979-1014. <https://www.jstor.org/stable/2937954>

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